

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0275

Meaux 138 kV
April 19, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0275. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the Meaux 138 kV substation in the Virginia area. The request was received on September 26, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 174 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$221,620,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

White Stone / Martn Dam 138.0 kV: 692.2% loading (211 MVA) following the outage of the Meaux / Hat Rapids 138.0 kV; DFAX = 100.0%.

White Stone / Meaux 138.0 kV: 663.7% loading (224 MVA) following the outage of the Hat Rapids / Meaux 138.0 kV; DFAX = 100.0%.

For the loss of Meaux / White Stone 138.0 kV, the Hat Rapids / Meaux 138.0 kV L loads to 650.3% of its 229 MVA rating. Project DFAX: 100.0%.

New Denver / Martin Dam 138.0 kV: 427.8% loading (249 MVA) following the outage of the Hat Rapids / Meaux 138.0 kV; DFAX = 67.5%.

For the loss of Meaux / White Stone 138.0 kV, the Hat Rapids / Cosmar 138.0 kV line loads to 288.1% of its 222 MVA rating. Project DFAX: 41.5%.

For the loss of Meaux / White Stone Substation 138.0 kV, the Magnolia / Toano 138.0 kV loads to 287.9% of its 255 MVA rating. Project DFAX: 42.3%.

The Greening / New Denver 138.0 kV L is overloaded to 283.5% of its 251 MVA emergency rating for the loss of the Hat Rapids / Meaux 138.0 kV. The project contributes 47.5% DFAX (712.6 MW) to this overload.

For the loss of Toano / Magnolia Substation 138.0 kV L, the Cosmar / Rush Creek Substation 138.0 kV Line loads to 269.3% of its 206 MVA rating. Project DFAX: 29.0%.

Hat Rapids / York Canyon Substation 138.0 kV: 260.5% loading (197 MVA) following the outage of the White Stone / Meaux 138.0 kV; DFAX = 32.6%.

Columbus West / Rush Creek 138.0 kV: 258.3% loading (211 MVA) following the outage of the Toano / Magnolia 138.0 kV; DFAX = 29.0%.

Columbus West / Oahe 138.0 kV L: 245.0% loading (217 MVA) following the outage of the Mgnolia / Toano 138.0 kV; DFAX = 29.0%.

For the loss of New Denver / Greening 138.0 kV, the Lamping Sta / Daleville 138.0 kV loads to 210.4% of its 230 MVA rating. Project DFAX: 25.2%.

For the loss of Hat Rapids / Cosmar 138.0 kV L, the Caliente / Hat Rapids 138.0 kV loads to 205.1% of its 216 MVA rating. Project DFAX: 35.5%.

Martin Dam / York Canyon 138.0 kV: 196.8% loading (199 MVA) following the outage of the Meaux / Hat Rapids 138.0 kV; DFAX = 32.5%.

Magnolia / Caliente 138.0 kV: 186.9% loading (226 MVA) following the outage of the Cosmar / Hat Rapids 138.0 kV; DFAX = 35.5%.

The Daleville / New Denver 138.0 kV Line is overloaded to 186.1% of its 224 MVA emergency rating for the loss of the New Denver / Greening 138.0 kV L. The project contributes 23.4% DFAX (351.3 MW) to this overload.

The Grahamsville / Hewitt Rd 138.0 kV Line is overloaded to 177.8% of its 189 MVA emergency rating for the loss of the Hat Rapids / Meaux 138.0 kV. The project contributes 18.4% DFAX (276.2 MW) to this overload.

Heitt Rd / Acec Lewiston 138.0 kV: 147.3% loading (199 MVA) following the outage of the New Denver / Martin Dam 138.0 kV line; DFAX = 11.0%.

The Wilkens / Highgrove Substation 138.0 kV is overloaded to 122.8% of its 307 MVA emergency rating for the loss of the Taber / Dixleta 345.0 kV. The project contributes 6.1% DFAX (90.8 MW) to this overload.

For the loss of Shingletown / Riva Rd 345.0 kV, the Dixleta / Taber 345.0 kV loads to 122.5% of its 1,108 MVA rating. Project DFAX: 19.3%.

The High Ridge / North Pole 138.0 kV line is overloaded to 120.9% of its 372 MVA emergency rating for the loss of the Dixleta / Taber 345.0 kV line. The project contributes 8.1% DFAX (121.5 MW) to this overload.

For the loss of Hat Rapids / Meaux 138.0 kV, the Zephylls / Lamping Station 138.0 kV loads to 119.9% of its 234 MVA rating. Project DFAX: 16.3%.

Grahamsville / Greening 138.0 kV: 119.3% loading (278 MVA) following the outage of the Daleville / Lamping Station 138.0 kV; DFAX = 12.4%.

Grant Street / Daleville 138.0 kV: 119.3% loading (198 MVA) following the outage of the New Denver / Martin Dam Substation 138.0 kV; DFAX = 10.5%.

Grant St / Binary 138.0 kV: 119.1% loading (204 MVA) following the outage of the New Denver / Martin Dam 138.0 kV line; DFAX = 10.5%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Martn Dam / White Stone 138.0 kV	Rebuild line with higher-capacity conductor	\$4,622,000	100% (\$4,622,000)
Meaux / White Tone Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$12,595,000	100% (\$12,595,000)
Hat Rapids / Meaux 138.0 kV L	Rebuild line with higher-capacity conductor	\$8,858,000	100% (\$8,858,000)
Martin Dam / New Denver 138.0 kV	Rebuild line with higher-capacity conductor	\$28,101,000	98% (\$27,538,000)
Cosmar / Hat Rapids 138.0 kV	Rebuild line with higher-capacity conductor	\$9,686,000	86% (\$8,333,000)
Magnolia / Toano 138.0 kV	Rebuild line with higher-capacity conductor	\$18,792,000	74% (\$13,884,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Greening / New Denver 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,214,000	97% (\$3,120,000)
Cosmar / Rush Creek 138.0 kV L	Rebuild line with higher-capacity conductor	\$10,347,000	63% (\$6,482,000)
York Canyon / Hat Rapids 138.0 kV	Rebuild line with higher-capacity conductor	\$35,443,000	70% (\$24,850,000)
Columbus West / Rush Creek 138.0 kV	Rebuild line with higher-capacity conductor	\$8,358,000	63% (\$5,236,000)
Oahe / Columbus W Substation 138.0 kV Line	Rebuild line with higher-capacity conductor	\$6,178,000	63% (\$3,870,000)
Lamping Sta / Daleville 138.0 kV	Rebuild line with higher-capacity conductor	\$5,373,000	70% (\$3,743,000)
Hat Rapids / Caliente Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$12,268,000	66% (\$8,049,000)
York Canyon / Martin Dam 138.0 kV	Rebuild line with higher-capacity conductor	\$29,620,000	96% (\$28,413,000)
Caliente / Magnolia 138.0 kV	Rebuild line with higher-capacity conductor	\$3,332,000	66% (\$2,186,000)
Daleville / New Denver 138.0 kV Line	Rebuild line with higher-capacity conductor	\$15,605,000	74% (\$11,491,000)
Hewitt Rd / Grahamsville Substation 138.0 kV L	Rebuild line with higher-capacity conductor	\$8,234,000	100% (\$8,234,000)
Heitt Rd / Acec Lewiston 138.0 kV	Rebuild line with higher-capacity conductor	\$20,128,000	61% (\$12,188,000)
Wilkins / Highgrove 138.0 kV line	Rebuild line with higher-capacity conductor	\$854,000	22% (\$188,000)
Taber / Dixleta 345.0 kV Line	Rebuild line with higher-capacity conductor	\$22,136,000	20% (\$4,428,000)
High Ridge / North Pole 138.0 kV line	Rebuild line with higher-capacity conductor	\$3,521,000	19% (\$665,000)
Zephyills / Lamping Station 138.0 kV	Rebuild line with higher-capacity conductor	\$9,785,000	100% (\$9,785,000)
Grahamsville / Greening 138.0 kV	Rebuild line with higher-capacity conductor	\$6,847,000	89% (\$6,063,000)
Grant Street / Daleville 138.0 kV	Rebuild line with higher-capacity conductor	\$4,352,000	86% (\$3,764,000)
Grant St / Binary 138.0 kV	Rebuild line with higher-capacity conductor	\$3,509,000	86% (\$3,035,000)
	Total allocated to project		\$221,620,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.